Vol 3 Chapter 8 — Hyperfine Splitting and the Lamb Shift

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Chapter 8 — Hyperfine Splitting and the Lamb Shift

Atomic spectroscopy has two famous fine-structure phenomena: hyperfine splitting (the splitting of atomic energy levels by nuclear magnetic-moment coupling) and the Lamb shift (the small QED-induced shift between $2s_{1/2}$ and $2p_{1/2}$ states of hydrogen, discovered by Lamb in 1947). Both involve α-suppressed contributions to atomic energy levels and are calculated to high precision in standard QED.

BST organizes these via the $\alpha^{\text{BST primary}}$ exponent pattern (Lyra T2476, May 22, 2026): transition matrix elements scale as α^k where k is a BST primary integer depending on the multipole order.

8.1 The exponent pattern

$$\alpha^k \quad \text{with} \quad k \in \{N_c \text{ primary integer at multipole order}\}.$$

For electric-dipole (E1) transitions: $k = 1$. For magnetic-dipole (M1) and electric-quadrupole (E2): $k = 3\ (= N_c)$. For higher multipoles: $k = 2L - 1$ where L is the multipole order, giving substrate-primary integer values.

8.2 Hyperfine splitting

The hyperfine splitting of hydrogen's ground state — the 21-cm line famous in radio astronomy — is dominated by the magnetic-dipole interaction between the proton and electron spins. Substrate prediction matches the measured value at the sub-percent level via the α^3 multipole structure.

8.3 The Lamb shift

The Lamb shift between $2s_{1/2}$ and $2p_{1/2}$ is a higher-order QED effect computed perturbatively in standard physics. BST predicts the shift via substrate-derived Welton-Bethe vacuum-fluctuation contributions, matching measurement at ~0.4%.

Multi-month work (Elie B6 derivation) refines the substrate-side Lamb-shift calculation toward parts-per-million precision.

8.4 What comes next

Chapter 9 — Atomic Spectroscopy — extends to multi-observable $\alpha^{\text{BST primary}}$ verification.

Where to look this up: T2476 $\alpha^{\text{BST primary}}$ exponent pattern. Welton 1948 Lamb-shift derivation. For standard atomic spectroscopy: Bransden and Joachain Chapter 5.